

ARRAYS

Q5) Letters Needed To Spell Numbers

DS	Session	Arrays	Question Information	Level 1 Challenge 25
Problem Description: For some reason, your school's football team has chosen to spell out the numbers on their jerseys instead of using the usual digits. Being great fans, you're going to be ready to cheer for your favorite players by bringing letter cards so you can spell out their number. Each fan has different favorites, so they each need to bring different sets of letters. The English spellings for the numbers 0 to 12 are: ZERO ONE TWO THREE FOUR FIVE SIX SEVEN EIGHT NINE TEN ELEVEN TWELVE Input Format: Read a set of integers from 0 to 12, separated by spaces, representing one fan's favorite players. The last integer will be 999, marking the end of the line. Output Format: Print the same numbers, then a period and a space. Then, in alphabetical order, print all the letters the fan needs to be able to spell any one of the jersey numbers provided				

CODE:

```
C main.c
1  #include <stdio.h>
2  #include <string.h>
3
4  const char *words[] = {
5      "ZERO", "ONE", "TWO", "THREE", "FOUR", "FIVE", "SIX",
6      "SEVEN", "EIGHT", "NINE", "TEN", "ELEVEN", "TWELVE"
7  };
8
9  int main(void) {
10     int nums[100];
11     int count = 0;
12     int val;
13
14     while (scanf("%d", &val) == 1 && val != 999) {
15         nums[count++] = val;
16     }
17
18     int req[26] = {0};
19
20     for (int i = 0; i < count; i++) {
21         int cur_freq[26] = {0};
22         const char *w = words[nums[i]];
23         for (int j = 0; w[j] != '\0'; j++) {
24             cur_freq[w[j] - 'A']++;
25         }
26         for (int c = 0; c < 26; c++) {
27             if (cur_freq[c] > req[c]) {
28                 req[c] = cur_freq[c];
29             }
30         }
31     }
32
33     for (int i = 0; i < count; i++) {
34         printf("%d ", nums[i]);
35     }
36     printf(". ");
37
38     for (int c = 0; c < 26; c++) {
39         for (int k = 0; k < req[c]; k++) {
40             putchar('A' + c);
41         }
42     }
43     printf("\n");
44
45     return 0;
46 }
```

OUTPUT:

```
PS C:\Users\CH.SC.U4CSE25255\Projects\test> ./main
1 2 999
1 2 . ENOTW
```